Open in app ↗

Medium Search Write

Angular : Why Virtual DOM can’t integrate with Angular ?

Angular can use a Virtual DOM, but it deliberately doesn’t — because Ivy’s compiled, fine-grained updates make a Virtual DOM unnecessary and often slower in Angular’s architecture.

Piyali Das 3 min read · Apr 8, 2026

1

🧠 1. It’s not “can’t” — it’s “doesn’t need to”

Angular already solves the same problem (efficient DOM updates) using:

- **Ivy compiled instructions (ɵɵ…)**
- **Change detection (Zone.js / Signals)**

👉 So adding Virtual DOM would be **duplicate work**

⚙️ 2. Core Architectural Difference

⚛️ React (Virtual DOM)

```
State change
   ↓
Create Virtual DOM
   ↓
Diff old vs new
   ↓
Update real DOM
```

🅰️ Angular (Ivy)

```
State change
   ↓
Run compiled instructions
   ↓
Update ONLY affected DOM node
```

👉 Angular already knows:

- What changed
- Where to update

No diffing needed.

🔥 3. Why Virtual DOM doesn’t fit Angular

❌ Reason 1: Angular already has precise change tracking

With Ivy:

```
⊖⊖textInterpolate(ctx.name);
```

👉 Angular directly updates **that exact text node**

If Virtual DOM is added:

```
Angular detects change
   ↓
Creates Virtual DOM ❌ extra step
   ↓
Diffing ❌ extra step
   ↓
Update DOM
```

👉 Adds overhead instead of removing it

❌ Reason 2: Angular templates are precompiled

React:

- JSX → runtime diffing needed

Angular:

- HTML → compiled to instructions

👉 Angular skips the “figure out what changed” step

❌ Reason 3: Performance trade-off

```
| Approach    | Cost                    |
| ----------- | ----------------------- |
| Virtual DOM | CPU (diffing)           |
| Angular Ivy | Minimal (direct update) |
```

👉 Angular chooses:

> Less CPU, more compile-time intelligence

❌ Reason 4: Memory overhead

Virtual DOM:

- Keeps a full tree in memory

Angular:

- Uses lightweight structures (LView/TView)

👉 Angular is more memory-efficient

❌ Reason 5: Change detection model mismatch

React:

- Pull-based (re-render everything → diff)

Angular:

- Push-based (update specific bindings)

👉 Virtual DOM works best with React’s model, not Angular’s

🧠 4. Angular already solved VDOM problems differently

```
| Problem        | React Solution | Angular Solution     |
| -------------- | -------------- | -------------------- |
| Detect changes | Re-render      | Change detection     |
| Find diff      | Virtual DOM    | Precompiled bindings |
| Update DOM     | Patch          | Direct update        |
```

⚡ 5. Signals (Angular 17+) makes this even stronger

With Signals:

```
count.set(5);
```

👉 Angular knows EXACTLY:

- Which component
- Which binding
- Which DOM node

Flow:

```
Signal update
   ↓
Targeted update
```

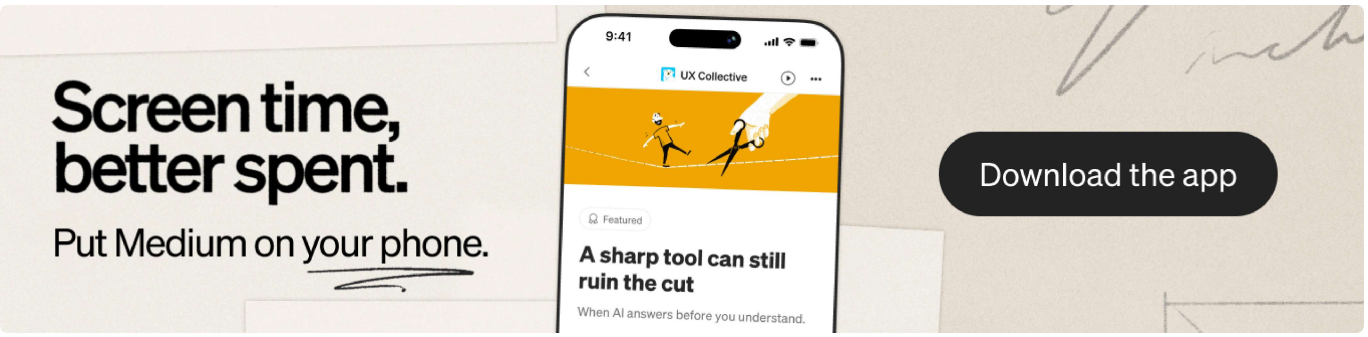
```
    ↓
DOM patch
```

👉 Even more precise than Virtual DOM

🔥 6. Could Angular support Virtual DOM?

Technically:

✓ Yes (you could build a layer)



But:

❌ No benefit
❌ Adds complexity
❌ Slower than Ivy

🧪 7. Real-world analogy

React:

> “Let’s redraw everything and compare”

Angular:

> “I already know what changed — just update that”

🧠 8. Deep Insight (important)

> Virtual DOM is a **runtime optimization**
>
> Ivy is a **compile-time optimization**

⚖️ 9. When Virtual DOM is better

React shines when:

- Dynamic UI generation
- Unknown structure at compile time
- Frequent full re-renders

⚡ 10. When Angular approach is better

Angular shines when:

- Structured apps
- Predictable templates
- Enterprise-scale systems

✅ Final Answer

👉 Angular doesn’t use Virtual DOM because:

- Ivy already provides **direct, precise updates**

- Virtual DOM would add **extra overhead**
- Angular relies on **compile-time intelligence instead of runtime diffing**

💡 One-line takeaway

> **React figures out changes at runtime**
> **Angular already knows changes at compile time**

Angular | Virtual Dom | Technology | Coding | UI

Written by Piyali Das

787 followers · 9 following

Edit profile

12 + yrs | Angular | NGRX | RxJS | https://www.linkedin.com/in/piyalidas10/
https://stackblitz.com/@piyalidas10 | https://github.com/piyalidas10

No responses yet

Piyali Das

What are your thoughts?